TruBridge Healthcare Data Analytics Project

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Topic: What drives cancer risk across Iowa's 99 counties, and what should TruBridge clients do about it?

BUSINESS CHALLENGE

Late-stage cancer costs 3-5x more to treat. Iowa's rural hospitals absorb that cost.

The financial hit

Stage IV colorectal cancer costs multiples of early-stage care. When patients show up too late — because they skipped screenings or couldn't afford care — rural hospitals absorb the difference.

TruBridge clients feel it

Rural and community health systems are TruBridge's core base. Thin margins mean late-stage cases hit revenue cycles hard. Preventable cases compound silently.

? The question no one is answering

Which Iowa counties are actually the highest-risk, and why? Without that data, health systems are guessing on where to direct screening outreach, insurance navigation, and prevention dollars.

THE APPROACH

Tested 3 social factors that are commonly thought to drive successful recovery outcomes from colorectal cancer.

01	**Poverty** *ACS Table S1701*	12 subgroups — overall poverty rate, child poverty, senior poverty, poverty by race, education level, and employment status across all 99 counties.
02	**Health Insurance** *ACS Table B27010*	Coverage type broken down by age group — private, employer-sponsored, Medicaid-only, and uninsured — separately for working-age adults (35–64) and seniors (65+).
03	**Rent Burden** *ACS Table B25070*	Share of renters paying more than 30% of income on housing (rent-burdened) and more than 50% (severely rent-burdened) across Iowa counties.

Outcome measured: Late-stage colorectal cancer incidence per 100K population & cancer risk probability

FACTOR 1 · POVERTY

Poverty is the most common assumption. Let's look at what the data shows.

Analyzed 12 poverty subgroups across all 99 Iowa counties.

10.9%

Median overall poverty rate

County range: 4%–22%

28.2%

Highest poverty group: Unemployed

vs 4.5% for employed

51 / 48

Counties split at median poverty

roughly even split

5

Counties missing Black poverty data

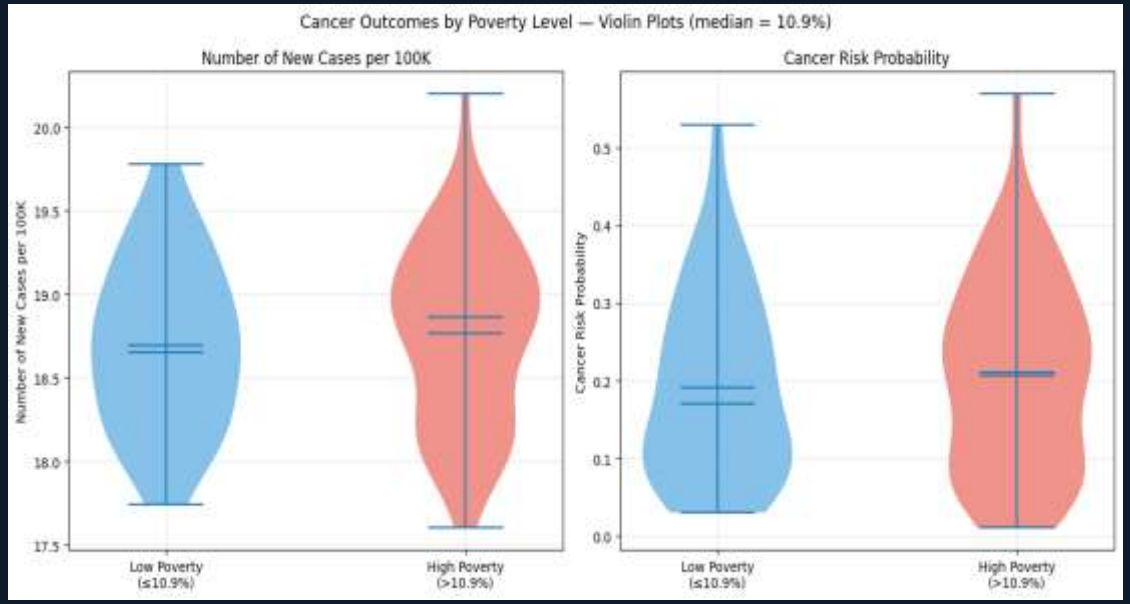
Sparse rural populations

The expectation going in: counties with more poverty would have higher cancer rates.
Higher poverty → less money for screenings → cancer caught later → higher incidence.

Hypothesis: If poverty drove cancer risk, splitting counties into high vs low poverty should show clearly different cancer outcomes. Let's see what happened.

FACTOR 1 · POVERTY — RESULT

High-poverty and low-poverty counties have essentially the same cancer outcomes.



What these charts show

Counties split at the median poverty rate (10.9%). The violin plots show the full distribution of cancer outcomes in each group. If poverty mattered, these shapes would look very different.

They don’t explain much.

The distributions overlap almost completely. High-poverty counties don't have meaningfully higher cancer rates. The maps of poverty and cancer risk point to completely different places.

Verdict: Poverty alone does ***not*** predict colorectal cancer risk in Iowa counties. Targeting cancer programs based on poverty rates would miss the counties that need them.

FACTOR 2 · HEALTH INSURANCE ACCESS

Insurance matters for cancer because it determines who gets screened.

Analyzed 5 insurance types (government, private, mixed insurance combinations) across working-age adults (35–64) and seniors (65+).

Screenings require coverage

Colonoscopies and other colorectal cancer screenings cost hundreds to thousands of dollars. Without insurance, most working-age adults simply skip them.

Uninsured delays = later diagnosis

Someone without coverage who notices symptoms often waits. By the time they seek care, the cancer has typically progressed further and is more expensive to treat.

Iowa's coverage gaps are real

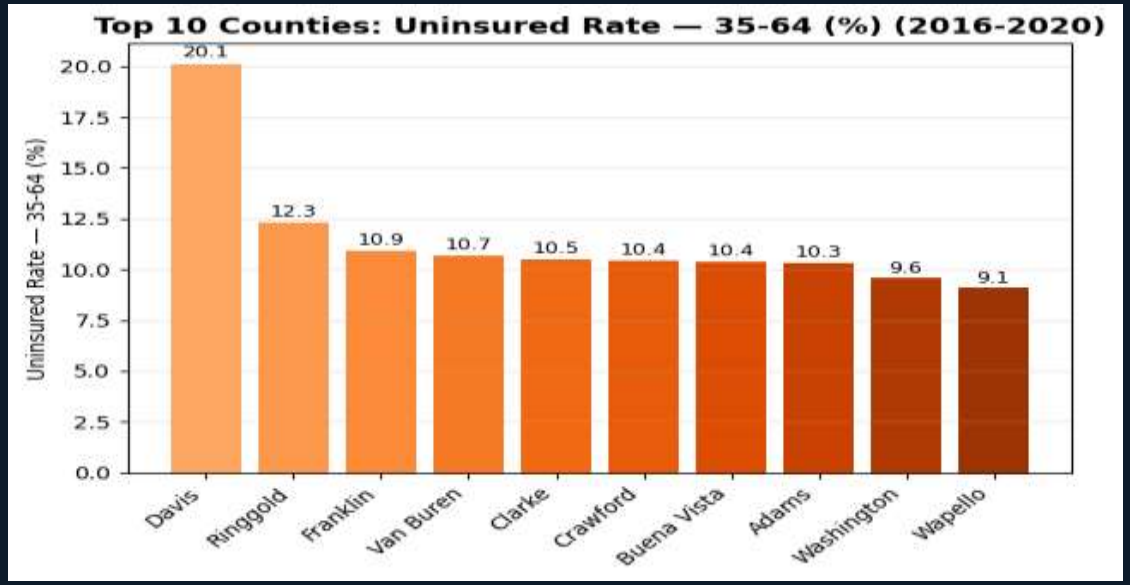
Uninsured rates among 35–64 year olds vary from under 2% to over 20% across Iowa counties — a massive range that maps directly onto healthcare access.

Hypothesis: Counties with higher uninsured rates have worse cancer outcomes because residents miss routine screenings that catch cancer early.

FACTOR 2 · INSURANCE ACCESS — RESULT

The most uninsured counties are not who you'd expect.

Top 10 counties by uninsured rate (working-age adults, 35–64):



Davis County: 20% uninsured

1 in 5 working-age adults in Davis County has no insurance — the highest in Iowa. These aren't the counties making headlines for poverty. They're rural counties where employer coverage is thin and Medicaid access is limited.

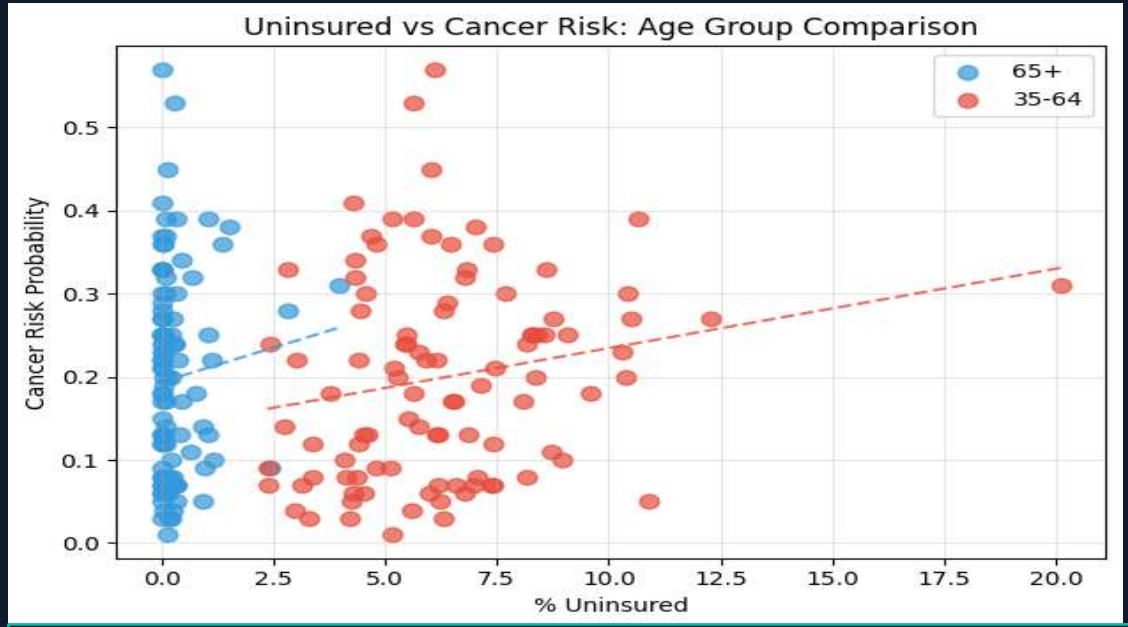
The pattern is consistent

High-uninsured counties are concentrated in rural Iowa. These are the same counties that show up with elevated cancer risk — not because they're poorer, but because their working-age adults have no coverage pathway to screenings.

Verdict: High uninsured rates cluster in rural counties where employer-sponsored coverage is thin and residents fall into coverage gaps — too poor for private plans, not poor enough for Medicaid.

FACTOR 2 · THE MEDICARE PROOF

When coverage becomes universal at 65, the cancer risk gap disappears entirely.



Ages 35-64 (no Medicare)

Each dot is a county. The upward trend for the red (35–64) dots shows clearly: the more uninsured working-age adults in a county, the higher the cancer risk. This is the coverage gap in action.

Ages 65+ (Medicare)

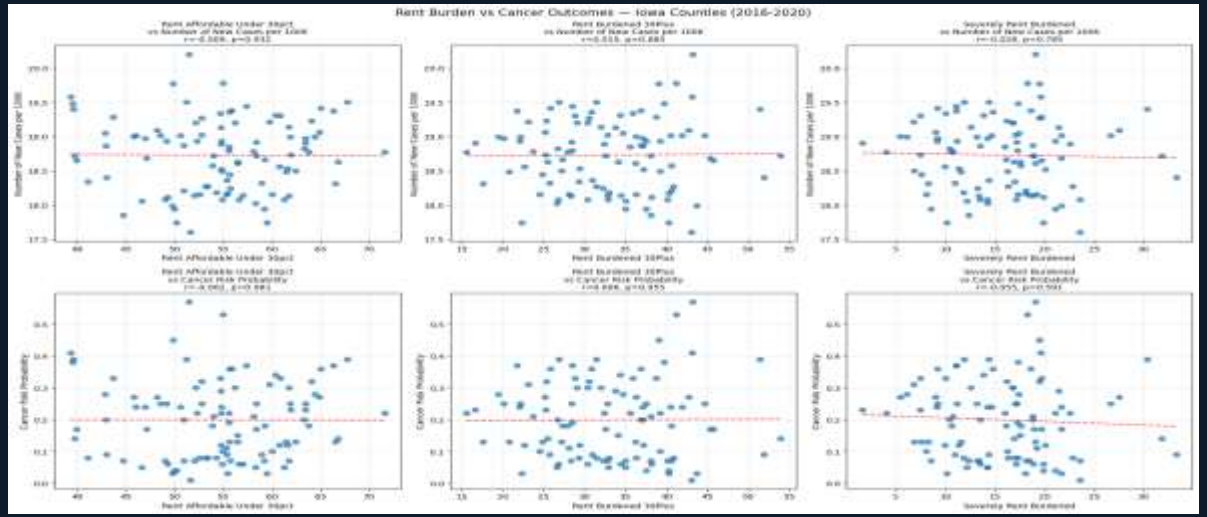
The blue dots (65+) are compressed near zero on the x-axis — nearly everyone is covered. The trend line is flat. The relationship between insurance and cancer risk vanishes because the coverage gap vanishes.

Verdict: This is the natural experiment. Medicare proves that universal coverage eliminates the disparity. Ages 35–64 is where the gap lives — and where intervention will have the highest impact.

FACTOR 3 · RENT BURDEN

Evaluated rent burden too. No relationship with cancer outcomes.

Every regression line is flat across all 6 combinations tested.



What was tested

- ✓ Affordable rent (<30% income)
- ✓ Rent burdened (30–50%)
- ✓ Severely burdened (50%+)

Vs. both outcomes:
→ Cancer cases per 100K
→ Cancer risk probability

Verdict: Iowa is relatively affordable compared to coastal states. Most counties have 50–60% of renters paying under 30% of income. The variation that exists doesn't track with cancer risk.

THE FULL PICTURE

Here's where each factor actually stands.

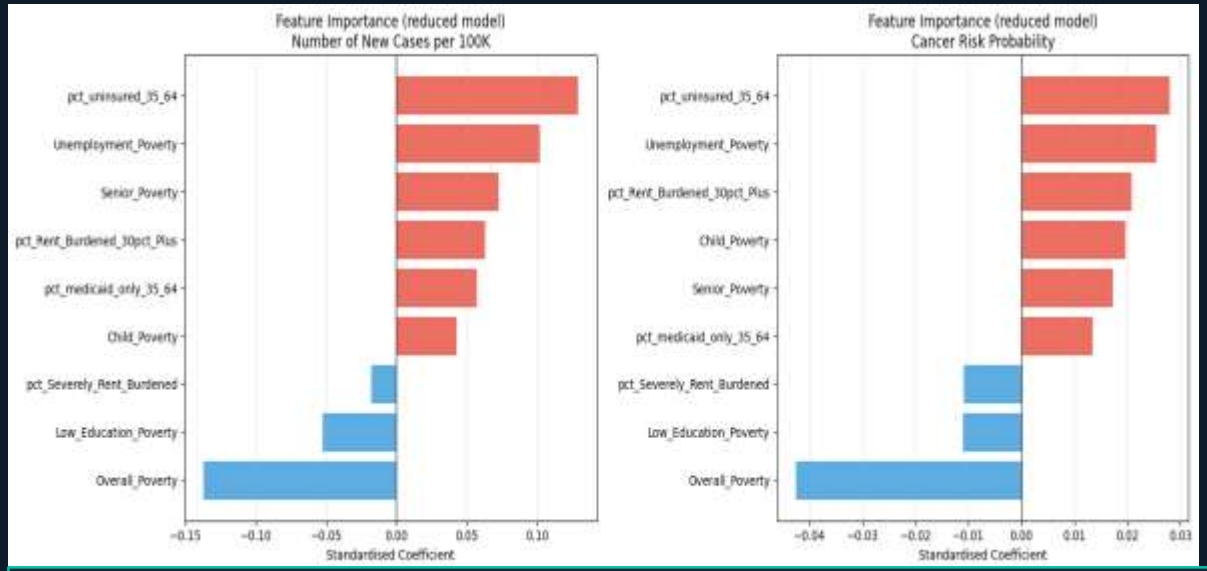
Private insurance coverage	Protective — counties with more private coverage have lower cancer risk	✓ Significant predictor
Uninsured rate (ages 35-64)	Risk factor — more uninsured adults in a county = higher cancer risk	✓ Significant predictor
Medicaid-only coverage	Elevated risk — counties with more Medicaid-only coverage show higher risk	✓ Significant predictor
Overall poverty rate	No meaningful relationship — high and low poverty counties look the same	✗ Not a predictor
Child & senior poverty	No meaningful relationship — tested all 12 subgroups, none were significant	✗ Not a predictor
Rent burden (all categories)	No relationship — flat lines across all 6 combinations tested	✗ Not a predictor

Key Takeaway: Insurance access is the only social factor that consistently and significantly predicts colorectal cancer risk across Iowa counties. Poverty and rent burden need intervention, but separately.

MULTIVARIATE ANALYSIS

When all 3 factors are combined into one model, Insurance access still outweighs others.

A regression and PCA analysis was run using all poverty, insurance, and rent variables together across 99 counties.



What this chart shows

Each bar is one social factor. Red bars raise cancer risk, blue bars lower it. The longer the bar, the stronger the effect. Uninsured rate is the longest red bar by far — it's the dominant driver even after controlling for everything else.

Model explains ~11% of variation

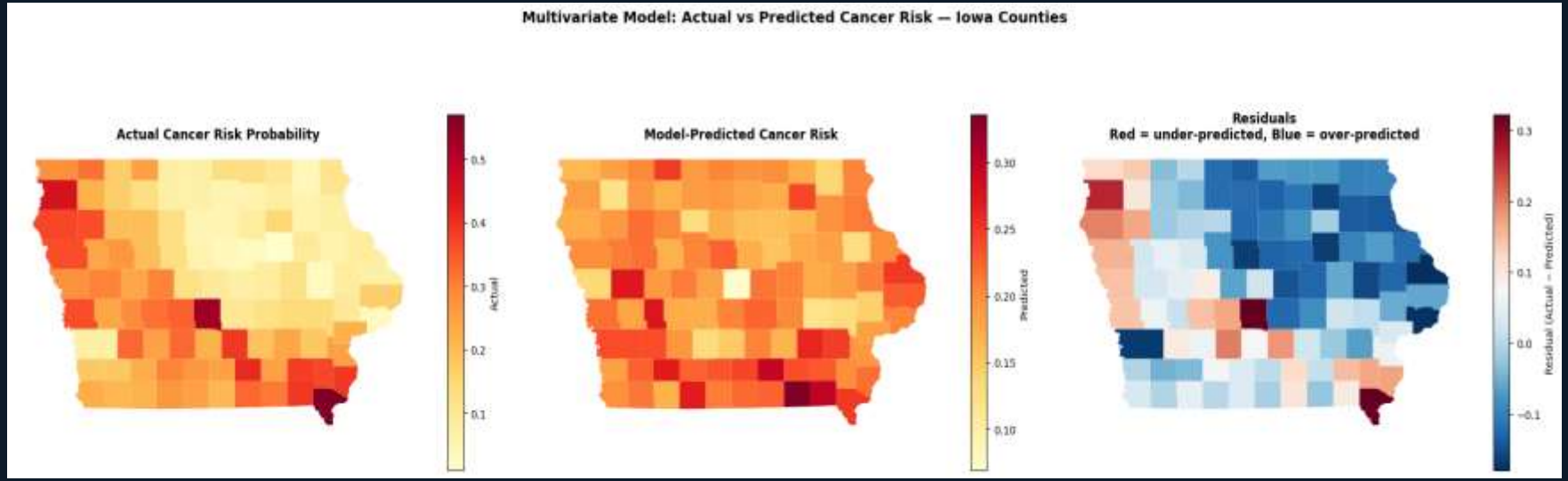
Expected at county level with 99 data points. Insurance + poverty + rent together explain ~11% of cancer risk. Rurality, diet, and screening uptake account for the rest.

Verdict: Even in a combined model with 9 predictors, uninsured rate among working-age adults is the single strongest positive predictor of cancer risk.

MULTIVARIATE ANALYSIS · MODEL RESIDUALS

The model gets most counties right. Red counties have unexplained high risk.

Actual cancer risk (left) vs what the SDOH model predicted (middle) vs the gap (right).



Where the model works well

Most of Iowa (blue residuals = slight over-prediction) fits the SDOH model. These counties' cancer risk is largely explained by their insurance and poverty profile.

Where risk is unexplained (red counties)

These counties have higher cancer risk than their SDOH profile predicts. Other factors like screening access, rurality, diet, are likely driving additional risk that insurance data alone can't capture.

TARGETED RECOMMENDATIONS

Where should TruBridge target?

Insurance Access, as insurance coverage for working-age adults (35–64) is the single social factor that predicts colorectal cancer risk across Iowa.

01 Target high-uninsured counties first

Counties like Davis, Ringgold, and Franklin — with 10–20% uninsured working-age adults — are the highest-yield targets for screening outreach and insurance navigation. Poverty maps would send you to different counties entirely.

02 Use Medicare as the policy benchmark

Near-zero uninsured at 65 makes the cancer risk gap disappear. That's the goal for working-age adults. This data gives TruBridge the evidence base to advocate for expanded insurance access with health system clients.

03 Keep cancer and poverty programs separate

Both problems are real and need resources — but they don't predict the same outcomes in the same counties. Bundling them dilutes both. Run separate, county-specific targeting strategies for each.

LIMITATIONS

What this analysis can and can't tell us.

Methodology

- **County-level data masks individual variation**: A county with 10% average poverty can still have high-risk pockets (ecological fallacy)
- **Correlation, not causation**: Uninsured rate tracks with cancer risk but we can't prove it directly causes higher rates without controlling for all confounders
- **$R^2 \approx 0.11$**: The model explains ~11% of cancer risk variation; 89% comes from factors not in this dataset

Data

- **ACS 5-year estimates carry margins of error:** Especially unreliable for small rural counties with sparse populations
- **2016–2020 snapshot**: Post-COVID Medicaid expansion and ACA enrollment shifts may have changed the coverage landscape since
- **Late-stage incidence only:** Measured cancer after it progressed, not screening rates which would be the more direct causal link

Missing Pieces

- **No rurality index:** Distance to screening facilities and oncologists is a likely driver not captured in ACS tables
- **No behavioral data:** Diet, obesity, smoking, and physical activity all affect colorectal cancer risk but weren't available at county level
- **No screening rate data:** Most direct mechanism between insurance and cancer outcomes; that was inferred but not directly measured

These limitations don't invalidate the finding; they define the scope. Insurance access as a cancer risk predictor is robust across multiple analysis methods. The gaps point to where future data collection should focus.

Feel free to explore the data yourself!

Built an interactive dashboard and Python notebooks where you can explore cancer risk and SDOH factors county by county across Iowa.

- **Interactive Dashboard:** https://claude.ai/public/artifacts/0e5ac430-e52c-4041-8eee-7122405756c2

- **GitHub Code Repository:** https://github.com/sriramprog/TruBridge-Healthcare-Data-Analytics